

Airframe Design - Optimization: Part II

Research Skills: Version Control

As part of this assignment, you will be expected to further develop introductory skills in version control. In order to do so, complete the following:

- a. Create at least 3 github issues (which should be based on the feedback you received in the previous assignment).
- b. Create a branch for this assignment.
- c. Add, Commit, and Push your assignment submission to your github repository.
- d. Open a pull request as a part of your submission for this assignment.

Technical Tasks: Basic Optimization

- a. Use an optimization package (such as SNOW.jl) to optimize the chord distribution to achieve the elliptic lift distribution. Specify the span to be 8.0 m, the pitch angle to be 5 degrees, and constraint the plane to provide 1.7 N of lift at a velocity of 1 m/s.
- b. Compare your chord distribution to an elliptic one. Give visual and numerical results.
- c. Explore the optimization. What happens if you change the lift constraint (change it to 10x the current one)? What happens if you change the tolerance of the optimization? What happens if you change the span?

Writing Tasks: Methodology and Results

As you now have a better handle on what you're doing, it would be best to flesh out the details of your methodology as well as formally outline the results section of your final report draft. For this assignment, complete the following writing tasks:

- a. Complete a full draft of your methodology section, including all of the information required to reproduce your work.
- b. Outline the results section of your report, including placeholders for figures and cues for discussion content to include.